



Build a Virtual Private Cloud (VPC)

JJ

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Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

NextWork VPC

IPv4 CIDR block Info

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy Info

Default



Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud is a service that allows users to create a virtual network within the Amazon Web Services (AWS) cloud. It's useful because it gives users complete control over their virtual network, including security and performance.

In today's project I created a VPC from scratch along with public subnets to use an internet gateway for public access.

Personal reflection

Not long at all. Roughly 20 minutes

I didn't expect how easy it was to setup the VPC, subnets and internet gateway.



Virtual Private Clouds (VPCs)

How VPCs work

VPCs are a private cloud computing environment that's hosted within a public cloud. VPCs provide a secure, isolated network within a public cloud that allows organizations to host resources without exposing them to the public internet.

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This allows me to create any necessary resources from day one that would require a VPC

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Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is Internet Protocol version 4, which is the most common way to write an IP address. IPv4 address are written as four sets of numbers separated by dots (e.g., 192.168.0.1).



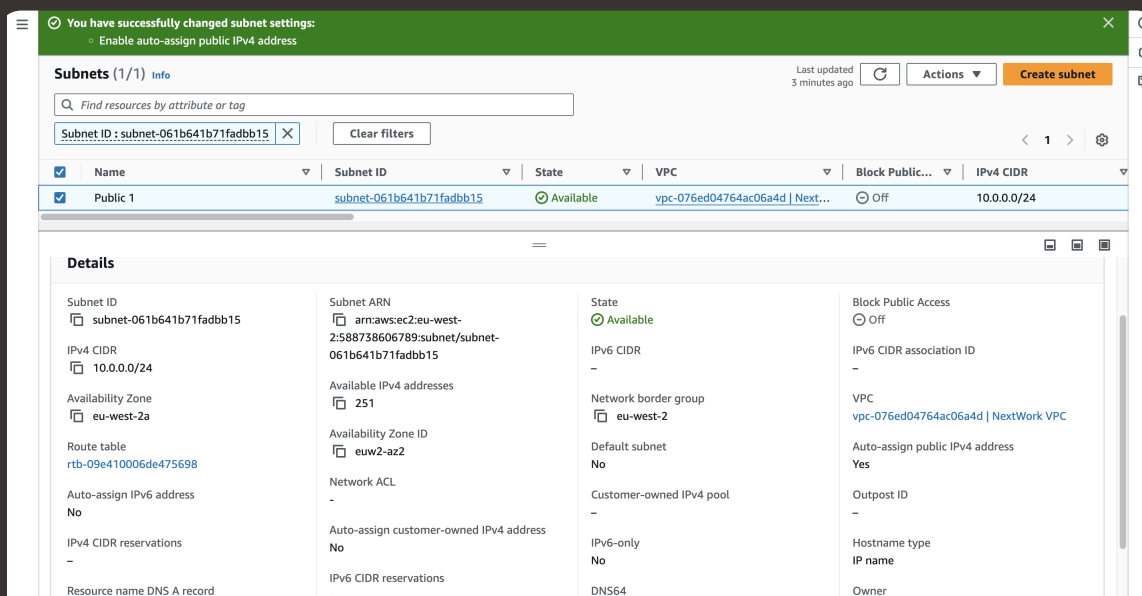
Subnets

Creating and configuring subnets

Subnets are a group resources with similar access rules and restrictions. Some subnets might be public areas that all resources can access (public subnets) while others are private areas with limited access (private subnets).

Public vs private subnets

The difference between public and private subnets are that with a public subnet, it is connected to the internet. (Resources inside a public subnet can communicate with external networks). Whereas a private subnet doesn't have access to the network





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Auto-assigning public IPv4 addresses

When you enable auto-assign public IPv4 address for a subnet, any EC2 instance launched in that subnet will instantly get a public IP address so you won't have to create one manually

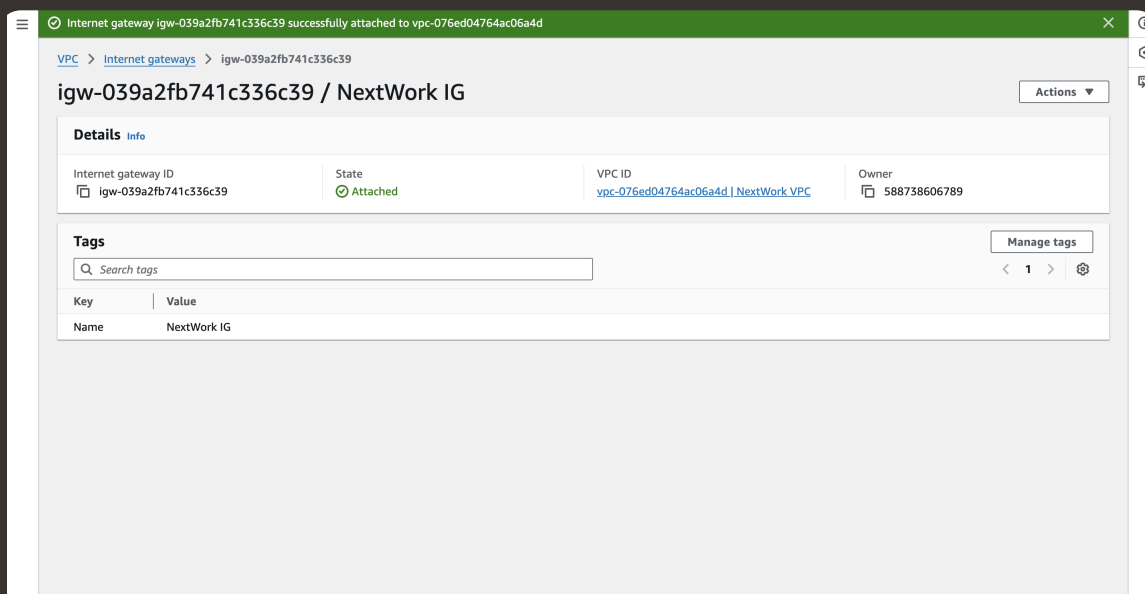


Internet gateways

Setting up internet gateways

An internet gateway connects your VPC and the outside world (internet). Internet gateways are key to making applications available on the internet. By attaching an internet gateway, your instances can access the internet.

Attaching an internet gateway to a VPC means resources in my VPC can now access the internet. The EC2 instances with public IP addresses also become accessible to users, so your applications hosted on those servers become public too.





Using the AWS CLI

Exploring CloudShell and CLI

VPC resources could also be created with CloudShell, which is a browser based command line tool built directly into the AWS console that lets you run commands without needing to install anything on your own computer. CLI stands for Command Line Interface, which is a way of interacting with AWS by typing text commands rather than clicking through the console, giving you more control and the ability to automate tasks.

Debugging my setup

To set up a VPC or a subnet, you can use the command line Make sure to avoid errors by including your VPC's CIDR block

```
aws
VPC > Your VPCs > vpc-0b413a8de91c1d3f0

VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud

Details
VPC ID vpc-0b413a8de91c1d3f0
State Available
Block Public Access Off
DNS hostnames Disabled
DNS resolution Enabled
Tenancy default
DHCP option set dopt-00cbb2a78db533722
Main route table rtb-0924d756065250ef6

CloudShell
eu-west-2
~ $ aws ec2 create-internet-gateway --query InternetGateway.InternetGatewayId --output text
igw-03b0e027aab5a9981
~ $ aws ec2 attach-internet-gateway --vpc-id vpc-0b413a8de91c1d3f0 --internet-gateway-id igw-03b0e027aab5a9981
~ $ aws ec2 attach-internet-gateway --vpc-id vpc-0b413a8de91c1d3f0 --internet-gateway-id igw-03b0e027aab5a9981

aws: [ERROR]: An error occurred (ResourceAlreadyAssociated) when calling the AttachInternetGateway operation: resource igw-03b0e027aab5a9981 is already attached to network vpc-0b413a8de91c1d3f0
~ $ aws ec2 create-internet-gateway --query InternetGateway.InternetGatewayId --output text
igw-0e5c81eb94013cf92
~ $ aws ec2 attach-internet-gateway --vpc-id vpc-0b413a8de91c1d3f0 --internet-gateway-id igw-0e5c81eb94013cf92

aws: [ERROR]: An error occurred (InvalidParameterValue) when calling the AttachInternetGateway operation: Network vpc-0b413a8de91c1d3f0 already has an internet gateway attached
~ $
```




Comparing CloudShell vs AWS Console

Compared to using the AWS Console, an advantage of using commands in CloudShell is that it is much faster once you know the commands, as you can create and attach resources in seconds without having to click through multiple menus and pages. An advantage of using the Console is that it is much more visual and beginner friendly, making it easier to understand what you are doing and where everything is as you go. Overall, I preferred the Console as someone new to AWS, since being able to see each resource being created made it much easier to follow along and understand what was happening at each step.



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